

CS & IT ENGINEERING



Digital Logic

Combinational Circuits

Lecture No. 05



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Recap of Previous Lecture



Topic

Set 1 }
Set 2 } mux

Started Set 3.



Topics to be Covered



Topic



set 3 }
set 4 }
✓ set 5

Set 3 :- Minimization.



(MUX + K-map)



exp



SCF (SOP)



K-map / properties
minimize.

(Q1)

Sol:-



$$y = \bar{P}\bar{Q} \cdot 1 + \bar{P}Q \cdot \bar{R} + P\bar{Q} \cdot 0 + PQ \cdot R$$
$$= \bar{P}\bar{Q} + \bar{P}Q\bar{R} + PQR$$

Step 2:- SCF (SOP)

$f(P, Q, R)$

$$y = \bar{P}\bar{Q} \cdot 1 + \bar{P}Q \cdot \bar{R} + P\bar{Q} \cdot 0 + PQ \cdot R$$

App 1:-

$$= \bar{P}\bar{Q} + \bar{P}Q\bar{R} + PQ R$$

$$= \bar{P} \cdot \bar{Q} (R + \bar{R}) + \bar{P}Q\bar{R} + PQ R$$

$$= [\bar{P}\bar{Q}R + \bar{P}\bar{Q}\bar{R} + \bar{P}Q\bar{R} + PQ R]$$

$$001 \quad 000 \quad 010 \quad 111$$

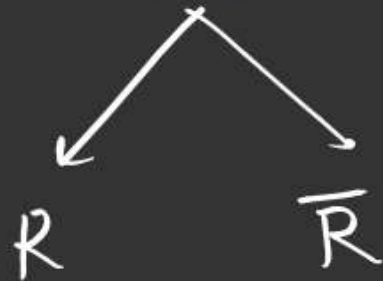
$$= \sum (0, 1, 2, 7) \checkmark$$

Step 2:- SCF (SOP)

$\underline{F(P, Q, R)}$

$$y = \bar{P}\bar{Q}.1 + \bar{P}Q.\bar{R} + P\bar{Q}.0 + PQ.R$$

$$= \bar{P}\bar{Q} + \bar{P}Q\bar{R} + PQ.R$$



App 2:-

$$PR\bar{Q} \Rightarrow \bar{P}\bar{Q}R$$

$$= \bar{P}\bar{Q}R + \bar{P}\bar{Q}\bar{R} + \bar{P}Q\bar{R} + PQ.R$$

$$= 001 \quad 000 \quad 010 \quad 111$$

$$= \Sigma(0, 1, 2, 7)$$

Step 2:- SCF (SOP)

$$\boxed{f(P, Q, R)}$$

$$y = \bar{P}\bar{Q} \cdot 1 + \bar{P}Q \cdot \bar{R} + P\bar{Q} \cdot 0 + PQ \cdot R$$

$$= \bar{P}\bar{Q} + \bar{P}Q\bar{R} + PQR$$



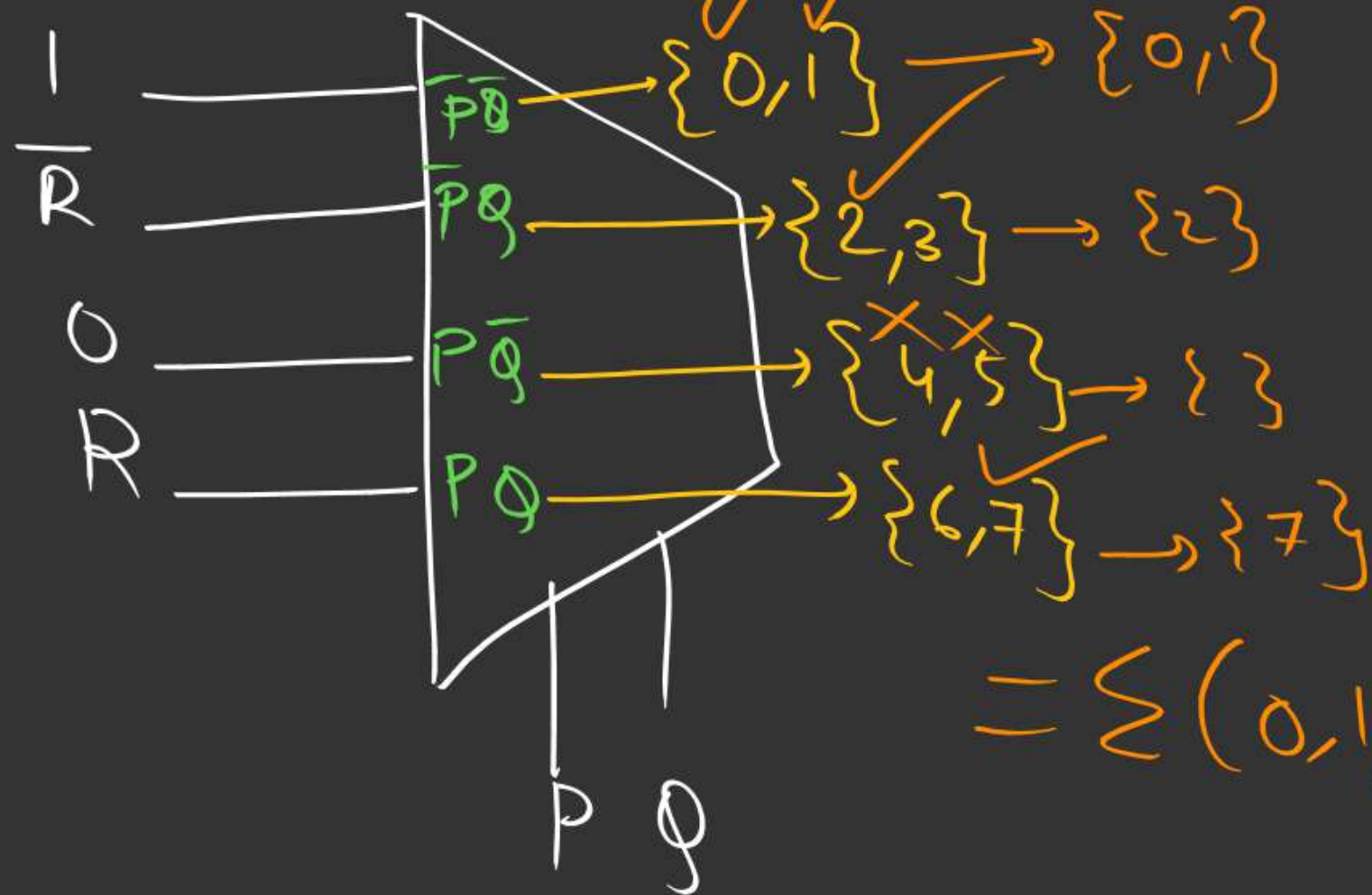
$$= 000, 001, 010, 111$$

$$= \sum m(0, 1, 2, 7)$$

App 3:-

$$\overline{P}\overline{Q} \cdot (1) \begin{matrix} \nearrow R \\ \searrow \overline{R} \end{matrix}$$

✓
App 9: (Direct shortcut)



PQ	R	
00	0	(0)
00	1	(1)
01	0	(2)
01	1	(3)
10	0	(4)
10	1	(5)
11	0	(6)
11	1	(7)



3 min / 9V



(Q)

$$F(P, Q, R) = ?$$

4x1 mux



$$\bar{P}\bar{Q}\bar{R} + \bar{P}Q \cdot 1 + P\bar{Q}R + PQ \cdot 1$$

$$= \bar{P}\bar{Q}\bar{R} + \bar{P}Q(R + \bar{R}) + P\bar{Q}R + PQ(R + \bar{R})$$

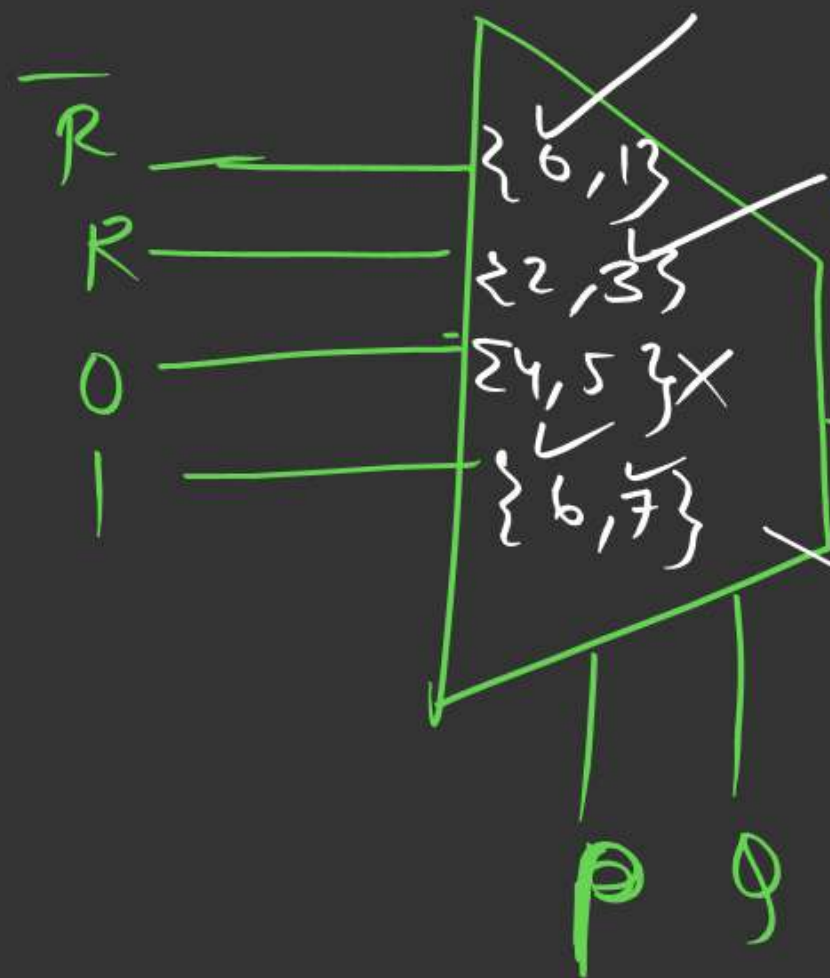
$$= \bar{P}\bar{Q}\bar{R} + \bar{P}QR + \bar{P}Q\bar{R} + P\bar{Q}R + PQR + PQ\bar{R}$$

$$= 000, 011, 010, 101, 111, 110$$

$$F(P, Q, R) = ? \quad \Sigma(0, 2, 3, 5, 6, 7)$$

$$= \Sigma(0, 2, 3, 5, 6, 7) \quad \checkmark$$

(Q)3) Find minimized $f(P, Q, R) = ?$



$f(P, Q, R)$

$$f = \bar{P}\bar{Q}\bar{R} + \bar{P}QR + P\bar{Q}\cdot 0 + P\cdot 0\cdot 1$$
$$= \bar{P}\bar{Q}\bar{R} + \bar{P}QR + P\cdot 0\bar{R} + P\bar{Q}\bar{R}$$

$$= 000, 011, 111, 110$$

$$\Rightarrow \sum m(0, 3, 6, 7)$$

$$\sum(0, 3, 6, 7) \checkmark$$

$\begin{array}{c} P \\ \swarrow \searrow \\ Q \quad R \end{array}$		$\bar{Q}\bar{R}$	$\bar{Q}R$	QR	$Q\bar{R}$
		0	1	3	2
$\bar{P}$	1		1		
P		4	5	1	6

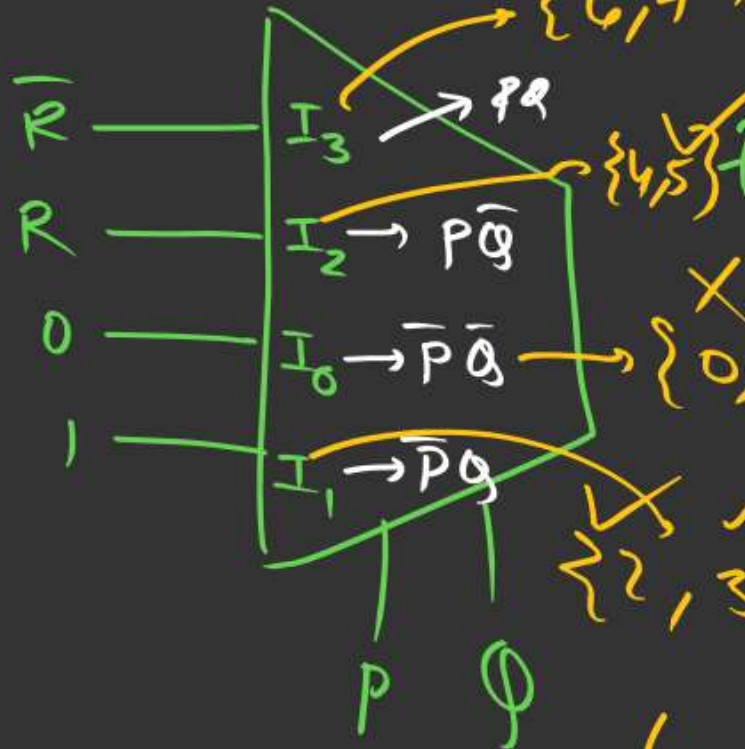
$\bar{P}\bar{Q}\bar{R}$ (points to cell 0)
 QR (points to cell 3)
 PQ (points to cell 6)

$$F(P, Q, R) = [\bar{P}\bar{Q}\bar{R} + QR + PQ]$$

$$(Q) f(P, Q, R) = ?$$

default

I_0
 I_1
 I_2
 I_3



$$f = PQ\bar{R} + P\bar{Q}R + \bar{P}\bar{Q} \cdot 0 + \bar{P}Q1$$

$$X = PQ\bar{R} + P\bar{Q}R + \bar{P}\bar{Q}R + \bar{P}Q\bar{R}$$

$$110 \quad 101 \quad 011 \quad 010$$

$$= \Sigma(2, 3, 5, 6)$$



$$\rightarrow \text{hottant} \Rightarrow \Sigma(2, 3, 5, 6)$$

$\begin{array}{c} \text{P} \\ \hline \text{Q} \end{array}$	$\overline{Q}R$	$\overline{Q}\overline{R}$	QR	$Q\overline{R}$
$\overline{P}$	0	1	3	2
P	4	5	7	6

$\overline{P}Q$ (points to cells 2 and 3)
 $P\overline{Q}R$ (points to cell 5)
 $Q\overline{R}$ (points to cells 2 and 6)

$$f(P, Q, R) = [P\overline{Q}R + Q\overline{R} + \overline{P}Q]$$

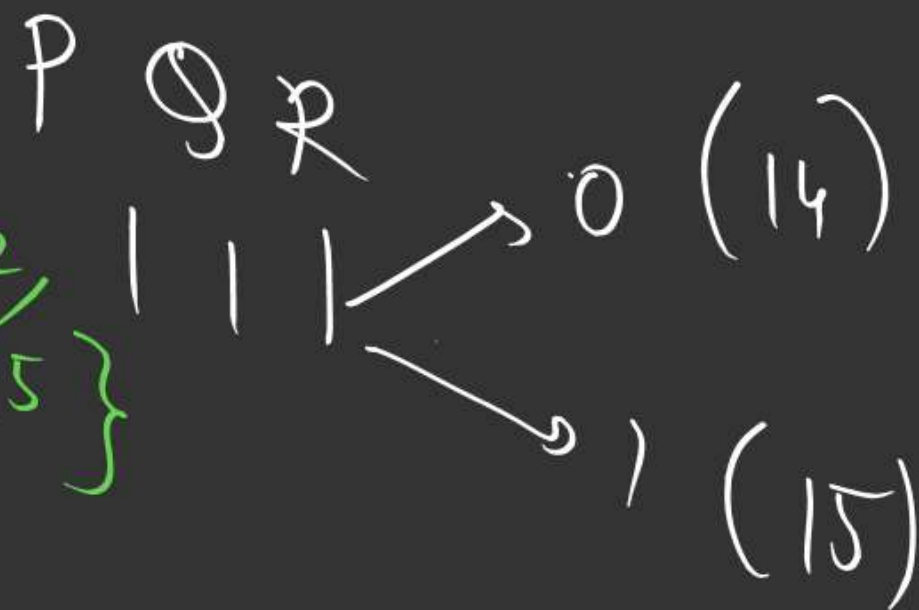
(Q) find minimized $f(P, Q, R, S) = ?$

S	—	{0, 1} $\bar{P}\bar{Q}\bar{R}$
0	—	{2, 3} $\bar{P}\bar{Q}R$
0	—	{4, 5} $\bar{P}Q\bar{R}$
1	—	{6, 7} $\bar{P}QR$
0	—	{8, 9} $P\bar{Q}\bar{R}$
1	—	{10, 11} $P\bar{Q}R$
0	—	{12, 13} PQR
1	—	{14, 15} PQR

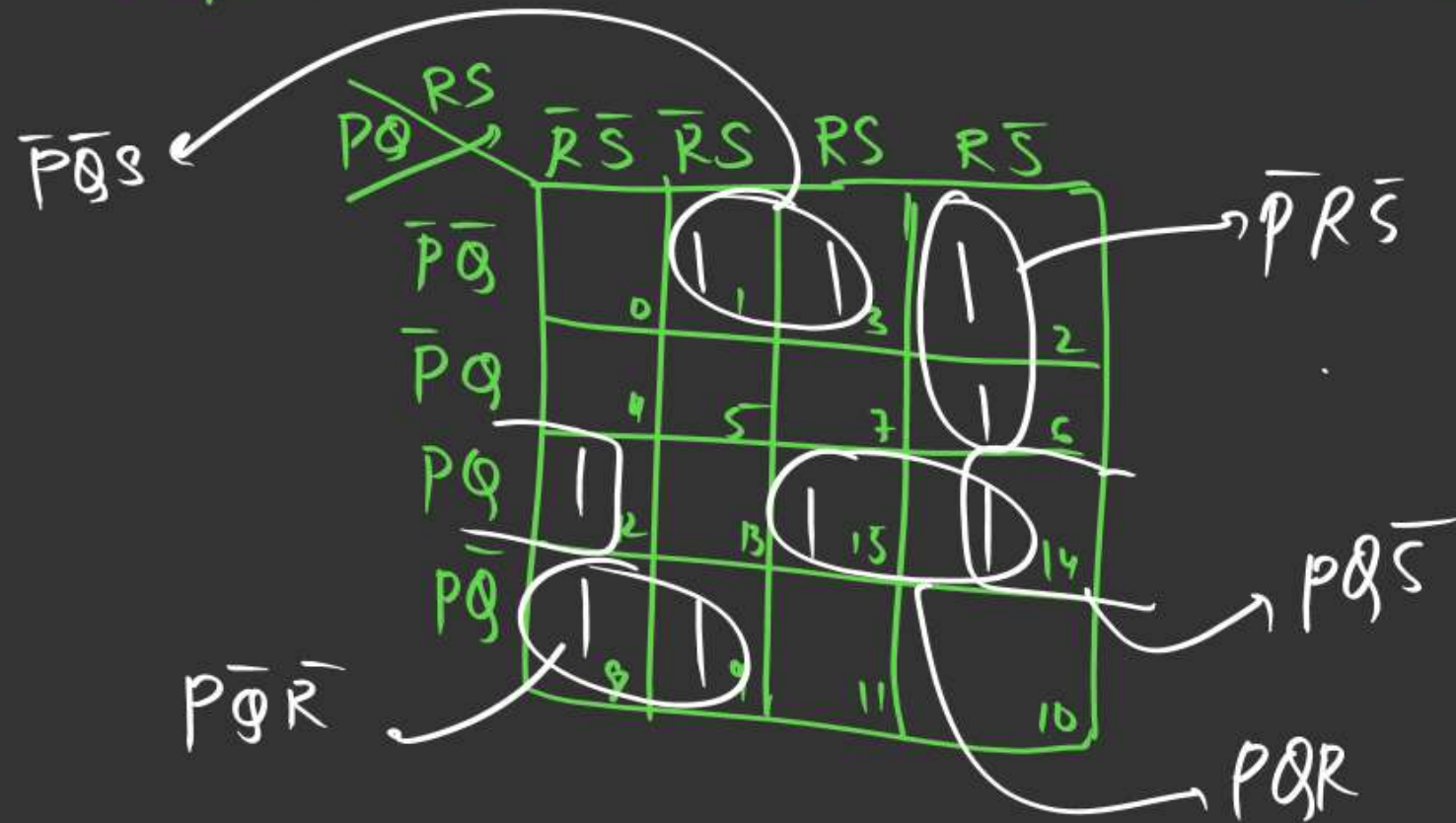
P Q R

$f(P, Q, R, S)$

$$= \Sigma(1, 2, 3, 6, 8, 9, 12, 14, 15)$$



Step 3: $f(p, q, r, s) = \sum (1, 2, 3, 6, 8, 9, 12, 14, 15)$



$\rightarrow F(p, q, r, s)$

$= \bar{P}R\bar{S} + PQ\bar{S} + PQR + P\bar{Q}\bar{R} + \bar{P}\bar{Q}S$

(g) find minimized $F(P, Q, R, S)$

P, Q, R, S

given



Order S, T_0

$f(P, Q, R, S)$

$$= R \cdot S \cdot P + R \bar{S} P Q + \bar{R} S P + \bar{R} \bar{S} (P + \bar{Q})$$

$$= PRS + PQ\bar{R}\bar{S} + P\bar{R}S + P\bar{R}\bar{S} + \bar{Q}\bar{R}\bar{S}$$

1 1 1 1 1 1 0 0

$$= P\bar{Q}\bar{R}S + P\bar{Q}RS + P\bar{Q}R\bar{S} + P\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}S$$

1011 (11) 1111 (15) 1110 (14) 1001 (9) 1101 (13)

$$+ P\bar{Q}\bar{R}\bar{S} + P\bar{Q}R\bar{S} + \bar{P}\bar{Q}\bar{R}\bar{S}$$

1000 (8) 1100 (12) 1000 (8)

$$= \sum (0, 8, 9, 11, 12, 13, 14, 15)$$

(mux + S(F) + kmap)

Step 2:

$PQ \backslash RS$	$\bar{R}\bar{S}$	$\bar{R}S$	RS	$R\bar{S}$
$\bar{P}\bar{Q}$	0	1	3	2
$\bar{P}Q$	4	5	7	6
PQ	12	13	15	14
$P\bar{Q}$	8	9	11	10

$\bar{Q}\bar{R}\bar{S}$ (points to cell 0)
 PS (points to cell 9)
 PQ (points to cell 14)

K-map

- 1) Large groups
 and
 2) lesser groups

$$F(P, Q, R, S) = [PS + PQ + \bar{Q}\bar{R}\bar{S}]$$

Remind → (1 variable ordering)

↓
Set 3 ✓

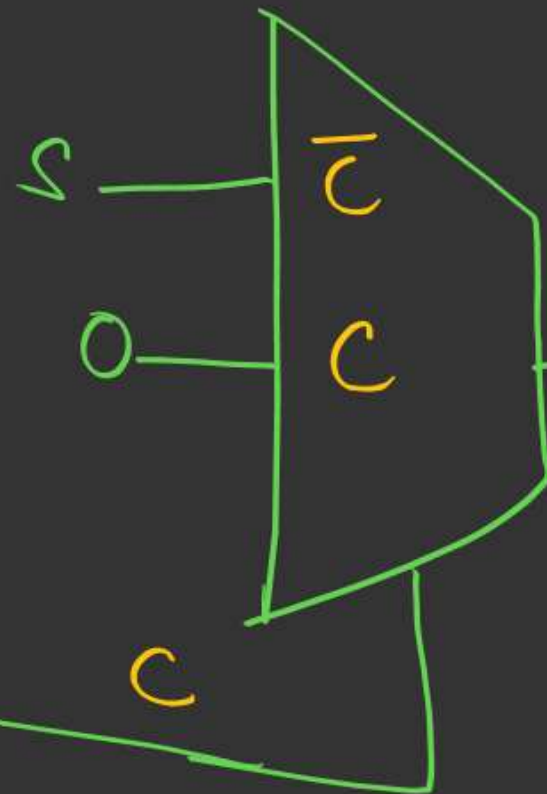
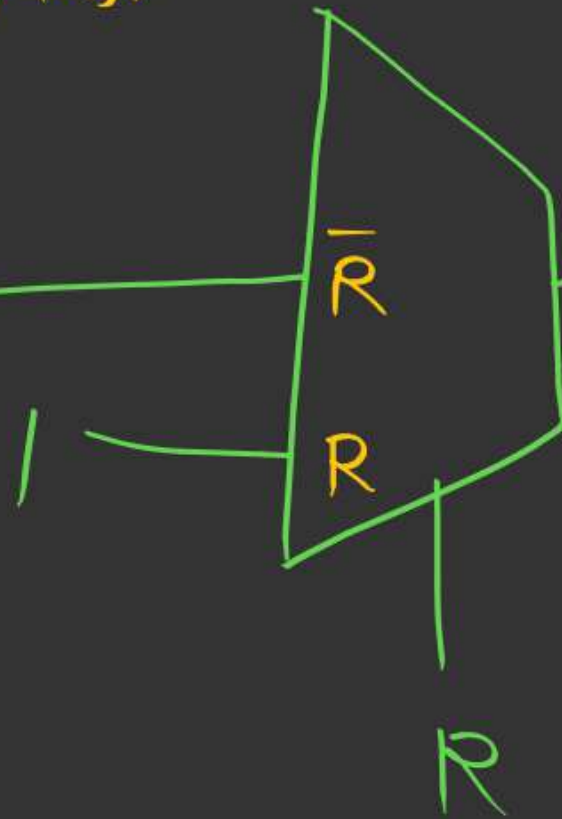
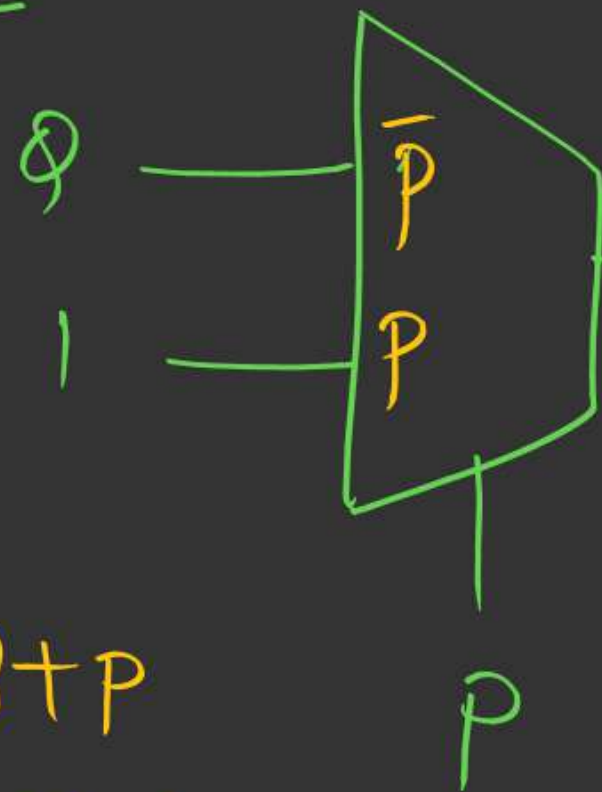
Set 4 Cascading of MUX (easier[↑])

$$\begin{aligned}
 b &= \bar{R} \cdot (P+Q) + R \cdot 1 \\
 &= \bar{R}P + \bar{R}Q + R \\
 &= P + R + \bar{R}Q \\
 &= P + R + Q
 \end{aligned}$$

$$y = f(P, Q, R, S)$$

$$\begin{aligned}
 a &= \bar{P}Q + P \cdot 1 \\
 &= (P+Q)
 \end{aligned}$$

eg 1)

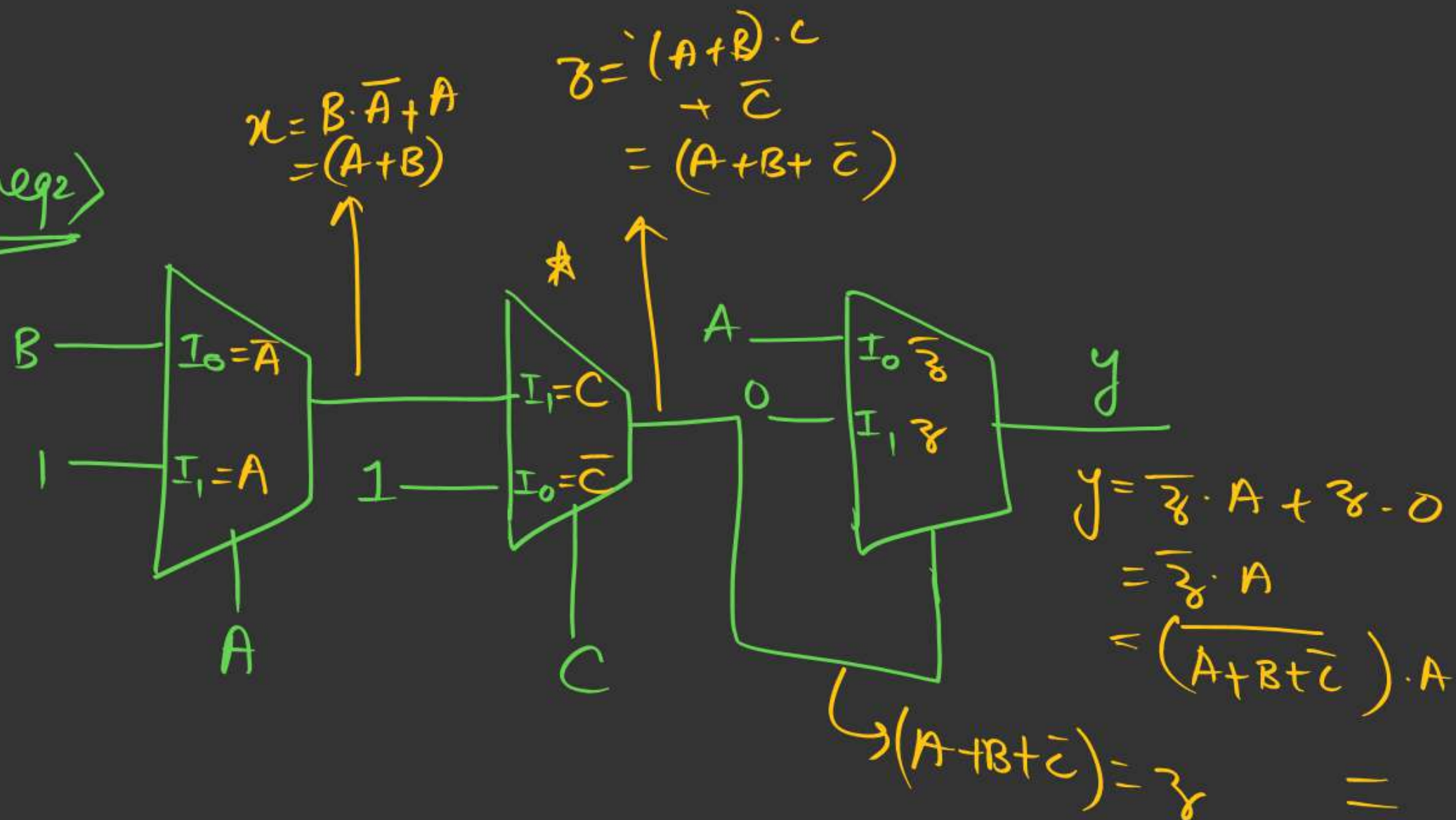


$y = ?$

$$\begin{aligned}
 a &= \bar{P}Q + P \\
 &= (P + \bar{P})(P + Q) \\
 &= \underline{P + Q}
 \end{aligned}$$

$$\begin{aligned}
 &\bar{C} \cdot S + C \cdot 0 \\
 &= \bar{C} \cdot S \\
 &= \overline{(P + R + Q)} \cdot S \\
 &= \boxed{\bar{P} \bar{R} \bar{Q} \cdot S}
 \end{aligned}$$

eq2



$$\begin{aligned} \overline{(A+B+\overline{C})} &= \overline{Z} \\ &= \overline{A} \cdot \overline{B} \cdot \overline{\overline{C}} \cdot A \\ &= \overline{A} \cdot \overline{B} \cdot C \cdot A = 0 \end{aligned}$$



2 mins Summary



Set 3 $\rightarrow$ 90%

Set 4 $\rightarrow$ 40%



THANK - YOU

